TEXORA · SESSIONAL NOTES · STUDY SHEET

Production of Different Types of Stitches

Forming 301, 401, 504 and 516 on their own machines, identifying each by structure and run-back behaviour, and verifying the lockstitch thread factor of 2.5 by unpicking a measured seam.

TE-3108 · APPAREL MANUFACTURING TECHNOLOGY-II SESSIONAL · STUDY 2 · LEVEL 3 TERM 1

1. Aim and place of the study

Aim: to produce and identify the working stitch types of ISO 4915 — 301 lockstitch, 401 chainstitch, 504 overedge and 516 safety stitch — and to measure the thread each consumes against the standard factors of TE-3107 Topic 3.

2. Materials, machine settings and specimens

Materials and machines: SNLS machine (301), chainstitch machine (401), overlock set for 504 and for 516; contrasting thread colours for needle and looper/bobbin so the structures can be identified; poplin pieces 30 × 10 cm; a 30 cm ruler and a seam ripper for the consumption test. **Settings:** density 4 stitches/cm on all machines; tensions balanced; record needle system and size for each machine.

Trouser measurement chart used throughout TE-3108 (men's basic trouser, garment values in cm): waist S 77 · **M 82** · L 87 · XL 92 (grade 5); hip 97 · **102** · 107 · 112 (grade 5); outseam (incl. waistband) 102 · **104** · 106 · 107; inseam 76 · **78** · 80 · 81; thigh 60 · **62** · 64 · 66 (grade 2); knee 44 · **46** · 48 · 50 (grade 2); hem (bottom) width 38 · **40** · 42 · 44 (grade 2); front rise 25 · **26** · 27 · 28. Size M is the working size unless the study states otherwise. Seam allowances, as in TE-2207/TE-2208: **1 cm** general, **1.5 cm** at side seams, waistband and other adjustment seams, **2.5 cm** at hems. The shirt values of TE-2207/2208 (size M chest 107, length 73, shoulder 46, sleeve 61.5, collar 40; grades 5/2/2/1.5/2 cm) are used unchanged wherever a shirt is the specimen.

3. Samples to produce and identify

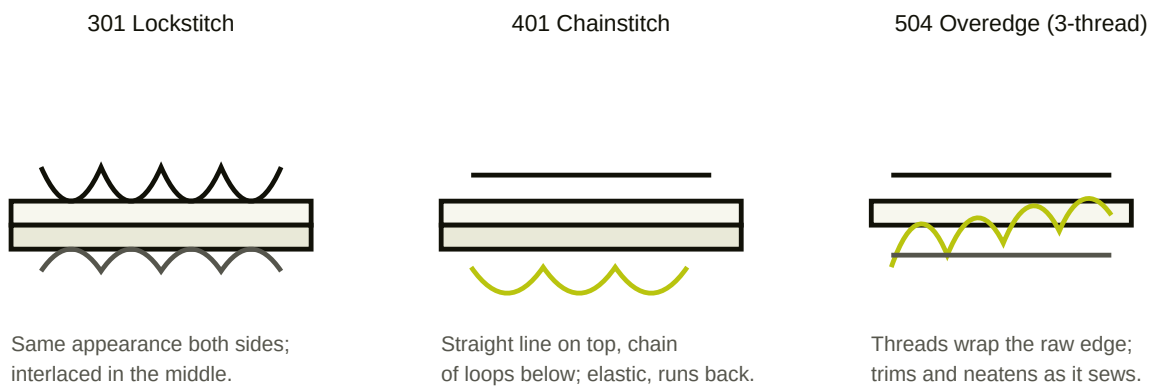


Figure 1 — The three workhorse stitch types of the sewing room (ISO 4915), top view above the fabric and underside below it.

STITCH	PRODUCE	IDENTIFICATION TEST
301 lockstitch	30 cm seam, two plies.	Same appearance both faces; pull a tail — the seam does not run back; unpick 10 cm and measure both threads.
401 chainstitch	30 cm seam, two plies.	Loops on the underside; pull the looper tail from the finishing end — the seam unchains.
504 overedge	30 cm along one raw edge.	Threads wrap the edge; the sample trims as it sews (note the trim width).
516 safety stitch	30 cm seam joining two plies with edge finish.	Two independent structures side by side: a 401 row inside a 504 row, sewn in one pass.

4. The thread-consumption measurement

Sew exactly 20 cm of 301 with marked threads. Unpick the seam completely, straighten (do not stretch) the recovered needle thread and bobbin thread, and measure each. **Expected result:** each thread measures about 25 cm, giving a ratio of 25 ÷ 20 = 1.25 per thread and **2.5 in total** — the standard factor of TE-3107 Topic 3, verified by your own hands. Record your measured values; a result between 2.3 and 2.7 confirms the method. Then compute, from the factors only, the thread in your 30 cm samples: 401: 0.30 m × 5.5 = 1.65 m; 504: 0.30 × 14.0 = 4.20 m; 516: 0.30 × 19.5 = 5.85 m — and state in your record why the overedge family consumes so much more (its threads wrap the edge and form wide loops at every stitch).

Observation points — what to record

- Face and underside appearance of each stitch, sketched or described in words.
- Run-back behaviour: which samples unravel from a tail, and from which end.

- Your measured 301 ratio against the standard 2.5, with the difference explained (density, thickness, unpicking loss).
- Trim width and edge quality of the 504 sample; independence of the two rows of the 516.

Precautions

- Mark the 20 cm seam length before sewing; an unmarked seam makes the consumption test meaningless.
- Unpick with the ripper under the threads, never by tearing — torn threads cannot be measured honestly.
- Note which tail you pull in the run-back test: a chainstitch unchains only from its finishing end.

Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Give the ISO class of each of the four stitches and the machine that forms it.
2. Your unpicked 301 threads measure 24 cm and 26 cm over a 20 cm seam. Calculate the total ratio and comment.
3. Why does the 516 sample remain a seam if its overedge row is cut away?
4. State the thread factor used in TE-3107 for 401 and the total thread in 12 m of 401 seam.

Answer key

1. 301 class 300 (lockstitch machine); 401 class 400 (chainstitch machine); 504 class 500 (overlock); 516 class 500 (overlock in safety-stitch configuration).
2. Total thread = 50 cm over 20 cm of seam: ratio 2.5 — exactly the standard factor; the small needle/bobbin difference reflects their slightly different paths around the interlacing.
3. Because 516 is two complete stitches sewn simultaneously: the 401 chain row alone still joins the plies (which is precisely the safety principle — the seam survives failure of the overedge row).
4. Factor 5.5 m per metre; 12 × 5.5 = 66 m of thread (needle + looper combined).
